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Previous Year Paper (Electronics) 18 Jan, 2025 Shift 1

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QUESTION PAPER
Electronics [SHIFT - 1]

Exam Date : 18/01/2025

Time : 03:00 PM - 05:00 PM

Part I - Aptitude

Question 1

Question Id : 126

Directions: *Study the following passage to answer the question that follows.*

Passage
Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, another hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not to take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

What happens when the deadline for a task gets extended unexpectedly?

Answer :

	Option Id
(A) The work is done more smoothly.	126001
(B) The work is done with more leisure.	126002
(C) The work consumes all the available time.	126003
(D) The work still needs additional time.	126004

Right Answer :
The work consumes all the available time.

Right Option Id : 126003

Question 2

Question Id : 127

Directions: *Study the following passage to answer the question that follows.*

Passage
Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, another hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not to take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

Explain the sentence: 'Work expands so as to fill the time available for its completion'.

Answer :

	Option Id
(A) The more work there is to be done, the more the time needed.	127001
(B) Whatever time is available for a given amount of work, all of it will get used.	127002
(C) If you have more time, you can do more work.	127003
(D) If you have some important work to do, you should always have some additional time.	127004

Right Answer :
Whatever time is available for a given amount of work all of it will get used.

Right Option Id : 127002

Question 3

Question Id : 128

Directions: *Study the following passage to answer the question that follows.*

Passage
Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, another hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not to take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

Who is the person likely to take more time to do the certain work?

Answer :

	Option Id
(A) A busy man	128001

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(B) A man of leisurely attitude
(C) An elderly person
(D) A relaxed man

128002
128003
128004

Right Answer :
A man of leisurely attitude

Right Option Id : 128002

Question 4

Question Id : 129

Directions: *Study the following passage to answer the question that follows.*

Passage
Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, another hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not to take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

What is the total time spent by the elderly lady in writing the postcard?

Answer : Option Id

(A) A full day — 129001
(B) Four hours and five minutes — 129002
(C) Well over half a day — 129003
(D) Seventy five minutes — 129004

Right Answer :
Seventy five minutes

Right Option Id : 129004

Question 5

Question Id : 130

Directions: *Study the following passage to answer the question that follows.*

Passage
Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, another hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not to take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

What does the expression 'pillar box' stand for?

Answer : Option Id

(A) A box attached to the pillar — 130001
(B) A box in the pillar — 130002
(C) Box office — 130003
(D) A pillar-type post box — 130004

Right Answer :
A pillar-type post box

Right Option Id : 130004

Question 6

Question Id : 131

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A cinema hall is 17 m 28 cm long and 11 m 52 cm broad. It is required to pave the floor of the hall by using the minimum number of square slabs of marble. How many such slabs are required?

Answer :

Option Id

(A) 4 — 131001
(B) 5 — 131002
(C) 6 — 131003
(D) 7 — 131004

Right Answer :
6

Right Option Id : 131003

Question 7

Question Id : 132

If the income of an employee is increasing annually by 10%, what will be the change in his income after two years?

Answer :

Option Id

(A) 21% increase — 132001
(B) 22% increase — 132002
(C) 23% increase — 132003
(D) 25% increase — 132004

Right Answer :
21% increase

Right Option Id : 132001

Question 8

Question Id : 133

Three years ago, the average age of a family of 5 members was 17 years. With the birth of a baby, the average age of six members remains the same even today. Find the age of the baby.

Answer :

Option Id

(A) 1 year — 133001
(B) 2 years — 133002
(C) 3 years — 133003
(D) 4 years — 133004

Right Answer :
2 years

Right Option Id : 133002

Question 9

Question Id : 134

A sum of Rs.8000 is deposited in the bank. The rate of interest in the bank for first year is 5% and for the second year it is 8%. Find the compound interest.

Answer :

Option Id

(A) Rs.1072 — 134001
(B) Rs.980 — 134002
(C) Rs.1124 — 134003
(D) Rs.1242 — 134004

Right Answer :
Rs.1072

Right Option Id : 134001

Question 10

Question Id : 135

A train 135 m long passes a boy standing on the platform in 7 seconds, but it passes the platform completely in 21 seconds. The length of the platform is _____.

Answer :

Option Id

(A) 310 m — 135001
(B) 185 m — 135002
(C) 215 m — 135003
(D) 270 m — 135004

Right Answer :
270 m

Right Option Id : 135004

Question 11

Question Id : 136

Find the odd one out.

ABC, BDF, CFI, DHL, EJM, FLR

Answer :

Option Id

(A) BDF — 136001
(B) DHL — 136002
(C) FLR — 136003
(D) EJM — 136004

Right Answer :
EJM

Right Option Id : 136004

Question 12

Question Id : 137

In a certain code, HIMALAYA is written as 201915271627327 and NILGIRI is written as 14191621191019. How is HARYALI written in that code?

Answer :

Option Id

(A) 2027103271619 — 137001
(B) 2010276271620 — 137002
(C) 2010296271619 — 137003
(D) 2011276271619 — 137004

Right Answer :
2027103271619

Right Option Id : 137001

Question 13

Question Id : 138

Aditi started travelling east from point A for 800 m. At the end of 800 m, she reached point B. She took a left turn from point B and travelled for another 500 m and reached point C. From point C, she turned left again and travelled for 400 m to reach point D. In which direction is point D from point A?

Answer :

Option Id

(A) North-East — 138001
(B) South-West — 138002
(C) North-West — 138003
(D) South-East — 138004

Right Answer :
North-East

Right Option Id : 138001

Question 14

Question Id : 139

Directions: *Study the following information carefully to answer the question that follows.*

Six friends A, B, C, D, E and F are sitting around a round table planning a carnival. They are not necessarily sitting in the same order. All are facing inside. A is sitting second to the left of B. F is sitting opposite to the person who sits to the immediate right of D. B sits between E on one side and F on the other. D doesn't sit between A and E. The friend sitting next to F doesn't sit to the immediate right of E.

Who among the following could be sitting between A and C?

Answer :

Option Id

(A) E — 139001
(B) B — 139002
(C) D — 139003
(D) Can't be determined — 139004

Right Answer :
D

Right Option Id : 139003

Question 15

Question Id : 140

Directions: *Study the following information carefully to answer the question that follows.*

Six friends A, B, C, D, E and F are sitting around a round table planning a carnival. They are not necessarily sitting in the same order. All are facing inside. A is sitting second to the left of B. F is sitting opposite to the person who sits to the immediate right of D. B sits between E on one side and F on the other. D doesn't sit between A and E. The friend sitting next to F doesn't sit to the immediate right of E.

Who is sitting to the left of F?

Answer :

Option Id

(A) A — 140001
(B) B — 140002
(C) C — 140003
(D) Can't be determined — 140004

Right Answer :
A

Right Option Id : 140001

Question 16

Question Id : 141

What is the name of combustible material at the tip of a safety match stick?

Answer :

Option Id

(A) Zinc — 141001
(B) Aluminium dioxide — 141002
(C) Nickel — 141003
(D) Antimony trisulphide — 141004

Right Answer :
Antimony trisulphide

Right Option Id : 141004

Question 17

Question Id : 142

Who among the following was the first Mughal emperor to allow Britishers to establish trade links with India?

Answer :

Option Id

(A) Akbar — 142001
(B) Jahangir — 142002
(C) Shah Jahan — 142003
(D) Aurangzeb — 142004

Right Answer :
Jahangir

Right Option Id : 142002

Question 18

Question Id : 143

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If any question arises whether a Bill is a Money Bill or not, the decision of the _____ thereon is final.

Answer :

Option Id

(A) Lok Sabha Speaker — 143001
(B) President of India — 143002
(C) Finance Minister of India — 143003
(D) None of these — 143004

Right Answer :
Lok Sabha Speaker

Right Option Id : 143001

Question 19

Question Id : 144

What do you understand by the term 'Mixed Economy'?

Answer :

Option Id

(A) Co-existence of small scale and large scale industries. — 144001
(B) Co-existence of the rich and the poor. — 144002
(C) Co-existence of private and public sector. — 144003
(D) Assigning equal importance to both agriculture and heavy industries. — 144004

Right Answer :
Co-existence of private and public sector.

Right Option Id : 144003

Question 20

Question Id : 145

Tides are the highest

Answer :

Option Id

(A) when the Earth is at its nearest to the Sun. — 145001
(B) when the Earth is at its nearest to the Moon. — 145002
(C) in the spring season. — 145003
(D) when the Sun, Moon and the Earth are in a straight line. — 145004

Right Answer :
when the Sun

Moon and the Earth are in a straight line.

Right Option Id : 145004

Question 21

Question Id : 146

Directions: *Study the following information to answer the question that follows.*

During mid-term selection examination events in a sports academy for three sports - Gymnastics, Football and Squash – 280 students appeared. When the results were declared, 185 students had passed in Gymnastics, 210 had passed in Football and 222 students had passed in Squash. Students who passed Football and Squash – 200. All those – except 5 students who passed in Gymnastics – passed in Football. All those – except 10 students who passed in Gymnastics – passed in Squash. 47 students failed in all 3 sports.

How many students passed in Squash only?

Answer :

Option Id

(A) 18 — 146001
(B) 21 — 146002
(C) 25 — 146003
(D) 29 — 146004

Right Answer :
18

Right Option Id : 146001

Question 22

Question Id : 147

Directions: *Study the following information to answer the question that follows.*

During mid-term selection examination events in a sports academy for three sports - Gymnastics, Football and Squash – 280 students appeared. When the results were declared, 185 students had passed in Gymnastics, 210 had passed in Football and 222 students had passed in Squash. Students who passed Football and Squash – 200. All those – except 5 students who passed in Gymnastics – passed in Football. All those – except 10 students who passed in Gymnastics – passed in Squash. 47 students failed in all 3 sports.

How many students passed in all 3 sports?

Answer :

Option Id

(A) 185 — 147001
(B) 175 — 147002
(C) 170 — 147003
(D) 171 — 147004

Right Answer :
171

Right Option Id : 147004

Question 23

Question Id : 148

Directions: *Study the following information to answer the question that follows.*

During mid-term selection examination events in a sports academy for three sports - Gymnastics, Football and Squash – 280 students appeared. When the results were declared, 185 students had passed in Gymnastics, 210 had passed in Football and 222 students had passed in Squash. Students who passed Football and Squash – 200. All those – except 5 students who passed in Gymnastics – passed in Football. All those – except 10 students who passed in Gymnastics – passed in Squash. 47 students failed in all 3 sports.

How many students failed in Football and Gymnastics?

Answer :

Option Id

(A) 65 — 148001
(B) 18 — 148002
(C) 58 — 148003
(D) 47 — 148004

Right Answer :
65

Right Option Id : 148001

Question 24

Question Id : 149

Directions: *Study the following information to answer the question that follows.*

During mid-term selection examination events in a sports academy for three sports - Gymnastics, Football and Squash – 280 students appeared. When the results were declared, 185 students had passed in Gymnastics, 210 had passed in Football and 222 students had passed in Squash. Students who passed Football and Squash – 200. All those – except 5 students who passed in Gymnastics – passed in Football. All those – except 10 students who passed in Gymnastics – passed in Squash. 47 students failed in all 3 sports.

How many students passed in Gymnastics but failed in other two sports?

Answer :

Option Id

(A) 5 — 149001
(B) 10 — 149002
(C) 15 — 149003
(D) 1 — 149004

Right Answer :
1

Right Option Id : 149004

Question 25

Question Id : 150

Directions: *Study the following information to answer the question that follows.*

During mid-term selection examination events in a sports academy for three sports - Gymnastics, Football and Squash – 280 students appeared. When the results were declared, 185 students had passed in Gymnastics, 210 had passed in Football and 222 students had passed in Squash. Students who passed Football and Squash – 200. All those – except 5 students who passed in Gymnastics – passed in Football. All those – except 10 students who passed in Gymnastics – passed in Squash. 47 students failed in all 3 sports.

How many students got promoted to next level of training, if they have to pass at least two sports?

Answer :

Option Id

(A) 180 — 150001

(B) 213 — 150002

(C) 200 — 150003

(D) 185 — 150004

Right Answer :

213

Right Option Id : 150002

Part II - Technical

Question 26

Question Id : 151

What term refers to the equivalent resistance of a capacitor, which depends on the frequency and the value of the capacitor?

Answer :

Option Id

(A) Impedance — 151001

(B) Resistance — 151002

(C) Reactance — 151003

(D) Conductance — 151004

Right Answer :

Reactance

Right Option Id : 151003

Question 27

Question Id : 152

What type of signal has values that are completely specific for any given time?

Answer :

Option Id

(A) Random Signal — 152001

(B) Deterministic — 152002

(C) Periodic Signal — 152003

(D) Aperiodic Signal — 152004

Right Answer :

Deterministic

Right Option Id : 152002

Question 28

Question Id : 153

Which of the following types of complements is used in digital systems for simplifying subtraction and logical manipulation, and is calculated by subtracting 1 from the base before finding the complement?

Answer :

Option Id

(A) Radix complement — 153001

(B) Diminished radix complement — 153002

(C) Two's complement — 153003

(D) One's complement — 153004

Right Answer :

Diminished radix complement

Right Option Id : 153002

Question 29

Question Id : 154

In 2's complement representation, what does a signed number with a 1 in the sign bit and all 0's in the magnitude bits represent?

Answer :

Option Id

(A) -2^n, where n is the number of bits in the magnitude — 154001

(B) $-2^{(n-1)}$, where n is the number of bits in the magnitude — 154002

(C) $-2^{(n+1)}$, where n is the number of bits in the magnitude — 154003

(D) -1, regardless of the number of bits — 154004

Right Answer :
-2^n where n is the number of bits in the magnitude

Right Option Id : 154001

Question 30

Question Id : 155

What is it called when the positive terminal of a battery is connected to the cathode and the negative terminal to the anode?

Answer :

Option Id

(A) Forward Biasing — 155001
(B) Neutral Biasing — 155002
(C) Zero Biasing — 155003
(D) Reverse Biasing — 155004

Right Answer :
Reverse Biasing

Right Option Id : 155004

Question 31

Question Id : 156

Which of the following codes is reflective, where the code for one digit is the complement of the code for its pair (e.g., 9 and 0, 8 and 1)?

Answer :

Option Id

(A) 8421 Code — 156001
(B) 2421 Code — 156002
(C) Excess-3 Code — 156003
(D) 5211 Code — 156004

Right Answer :
Excess-3 Code

Right Option Id : 156003

Question 32

Question Id : 157

What is the process of efficiently converting the output of a source into a sequence of binary digits called?

Answer :

Option Id

(A) Source Encoder — 157001
(B) Channel Encoder — 157002
(C) Data Modulation — 157003
(D) Signal Amplification — 157004

Right Answer :
Source Encoder

Right Option Id : 157001

Question 33

Question Id : 158

Which layer of the OSI model manages data packetization and delivery while checking for errors?

Answer :

Option Id

(A) Network Layer — 158001
(B) Data Link Layer — 158002
(C) Transport Layer — 158003
(D) Application Layer — 158004

Right Answer :
Transport Layer

Right Option Id : 158003

Question 34

Question Id : 159

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Which of the following is a weighted, sequential code used for mathematical operations, where each decimal digit is represented by a 4-bit binary number?

Answer :

Option Id

(A) 8421 BCD code — 159001
(B) Excess-3 code — 159002
(C) 2421 code — 159003
(D) Gray code — 159004

Right Answer :
8421 BCD code

Right Option Id : 159001

Question 35

Question Id : 160

What are hard real-time systems characterized by?

Answer :

Option Id

(A) Delayed task execution with minor consequences — 160001
(B) Execution of tasks without any specific deadlines — 160002
(C) Missing a task deadline can result in catastrophic consequences — 160003
(D) Tasks executed in a sequential manner without timing constraints — 160004

Right Answer :
Missing a task deadline can result in catastrophic consequences

Right Option Id : 160003

Question 36

Question Id : 161

Which code involves transmitting four data bits along with three parity bits, resulting in a 7-bit codeword?

Answer :

Option Id

(A) 8-bit Hamming code — 161001
(B) 7-bit Hamming code — 161002
(C) Parity bit code — 161003
(D) ASCII code — 161004

Right Answer :
7-bit Hamming code

Right Option Id : 161002

Question 37

Question Id : 162

What is the primary function of a Mobile Telephone Switching Office (MTSO) in a cellular network?

Answer :

Option Id

(A) To manage call routing between cell towers — 162001
(B) To serve as the central coordinating element for all cell sites — 162002
(C) To provide Internet connectivity to mobile phones — 162003
(D) To store customer data and provide email services — 162004

Right Answer :
To serve as the central coordinating element for all cell sites

Right Option Id : 162002

Question 38

Question Id : 163

What is the formula for the voltage gain (AV) of a Common Emitter Amplifier (CEA) at high input frequencies when the bypass capacitor C2 shorts the emitter branch to the ground?

Answer :

Option Id

(A) AV = (RC //RL) / re — 163001
(B) AV = (RC // RL) / (RE + re) — 163002
(C) AV = RC / RL — 163003
(D) AV = re / RC — 163004

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Right Answer :
AV = (RC //RL) / re

Right Option Id : 163001

Question 39

Question Id : 164

What is the formula for the bandwidth of a band-pass filter (BPF)?

Answer :

Option Id

(A) BW = ωc — 164001
(B) BW = ω2 - ω1 — 164002
(C) BW = ω1 + ω2 — 164003
(D) BW = ω1 / ω2 — 164004

Right Answer :
BW = ω2 - ω1

Right Option Id : 164002

Question 40

Question Id : 165

What does it mean for a set to be closed with respect to a binary operator?

Answer :

Option Id

(A) Every pair of elements from the set produces a unique result within the set. — 165001
(B) Every pair of elements from the set produces an element outside the set. — 165002
(C) The set only contains even numbers. — 165003
(D) The binary operator can only be applied to elements of the set. — 165004

Right Answer :
Every pair of elements from the set produces a unique result within the set.

Right Option Id : 165001

Question 41

Question Id : 166

Which of the following expressions represents one of DeMorgan's Theorems?

Answer :

Option Id

(A) (a + b)' = a'b — 166001
(B) (a * b)' = a' + b' — 166002
(C) a + (b' * c) = (a + b') * (a + c) — 166003
(D) (a * b) = a + b — 166004

Right Answer :
(a + b)' = a'b

Right Option Id : 166001

Question 42

Question Id : 167

What is the circuit called that converts AC into pulsating DC using a single diode?

Answer :

Option Id

(A) Half wave Rectifier — 167001
(B) Full wave Rectifier — 167002
(C) Bridge Rectifier — 167003
(D) Voltage Regulator — 167004

Right Answer :
Half wave Rectifier

Right Option Id : 167001

Question 43

Question Id : 168

What is the primary role of an oscillator in a microcontroller?

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Answer :

(A) To regulate the voltage levels in the system
(B) To provide a stable clock signal for controller operation
(C) To manage memory operations
(D) To process input signals

Option Id
168001
168002
168003
168004

Right Answer :
To provide a stable clock signal for controller operation

Right Option Id : 168002

Question 44

Question Id : 169

What converts a binary sequence into electrical signals for transmission over a channel?

Answer :

(A) Source Encoder
(B) Channel Decoder
(C) Signal Amplifier
(D) Digital Modulator

Option Id
169001
169002
169003
169004

Right Answer :
Digital Modulator

Right Option Id : 169004

Question 45

Question Id : 170

What does the Address Resolution Protocol (ARP) resolve?

Answer :

(A) Network Interface layer address to Internet layer address
(B) Internet layer address to Network Interface layer address
(C) Application layer address to Transport layer address
(D) Internet layer address to Application layer address

Option Id
170001
170002
170003
170004

Right Answer :
Internet layer address to Network Interface layer address

Right Option Id : 170002

Question 46

Question Id : 171

Which pins are used for serial data transmission and reception in a microcontroller's serial port?

Answer :

(A) TXD for reception and RXD for transmission
(B) TXD for transmission and RXD for reception
(C) SBUF for both transmission and reception
(D) Port 1 for transmission and reception

Option Id
171001
171002
171003
171004

Right Answer :
TXD for transmission and RXD for reception

Right Option Id : 171002

Question 47

Question Id : 172

What type of system schedules activities to follow a statically computed schedule?

Answer :

(A) Event-triggered
(B) Hard real-time
(C) Soft real-time
(D) Time-triggered

Option Id
172001
172002
172003
172004

Right Answer :
Time-triggered

Right Option Id : 172004

Question 48

Question Id : 173

What is the function of pin 30 (ALE) in the 8051 microcontroller?

Answer :

	Option Id
(A) It enables or disables external memory interfacing for the microcontroller.	173001
(B) It de-multiplexes the address and data signals of Port 0 for external memory interfacing.	173002
(C) It supplies +5V DC to the microcontroller for power.	173003
(D) It acts as a general-purpose input/output pin for user applications.	173004

Right Answer :

It de-multiplexes the address and data signals of Port 0 for external memory interfacing.

Right Option Id : 173002

Question 49

Question Id : 174

What causes severe fading in a mobile communication system?

Answer :

	Option Id
(A) Higher antenna height	174001
(B) Multipath waves generated by surrounding structures	174002
(C) Increased carrier frequency wavelength	174003
(D) Decreased antenna height and smaller carrier wavelength	174004

Right Answer :

Decreased antenna height and smaller carrier wavelength

Right Option Id : 174004

Question 50

Question Id : 175

Which theorem converts the internal capacitor CBC into the equivalent input and output capacitors Cin and Cout in a Common Emitter Amplifier (CEA)?

Answer :

	Option Id
(A) Miller's Theorem	175001
(B) Thevenin's Theorem	175002
(C) Norton's Theorem	175003
(D) Superposition Theorem	175004

Right Answer :

Miller's Theorem

Right Option Id : 175001

Question 51

Question Id : 176

What is the magnitude spectrum of a signal x(t) in the Fourier Transform?

Answer :

	Option Id
(A) $\|X(\omega)\|$	176001
(B) $\phi(\omega)$	176002
(C) $X(\omega) * e^{(j\phi(\omega))}$	176003
(D) $X(\omega)$	176004

Right Answer :

$|X(\omega)|$

Right Option Id : 176001

Question 52

Question Id : 177

Which section of the 8051 microcontroller contains bit-addressable locations where each bit has a unique address from 20H to 2FH?

Answer :

Option Id

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(A) General-purpose RAM
(B) Special Function Registers (SFRs)
(C) Bit-addressable RAM from 20H to 2FH
(D) Register Banks from 00H to 1FH

177001
177002
177003
177004

Right Answer :
Bit-addressable RAM from 20H to 2FH

Right Option Id : 177003

Question 53

Question Id : 178

What is the name of the eight-bit register in the 8051 microcontroller that provides the status of ALU operations and selects the register bank?

Answer :

Option Id

(A) Program Status Word (PSW) 178001
(B) Accumulator (A) 178002
(C) Stack Pointer (SP) 178003
(D) Data Pointer (DPTR) 178004

Right Answer :
Program Status Word (PSW)

Right Option Id : 178001

Question 54

Question Id : 179

What is the maximum reverse voltage a diode can withstand under reverse biasing called?

Answer :

Option Id

(A) Reverse Breakdown Voltage 179001
(B) Peak Inverse Voltage 179002
(C) Threshold Voltage 179003
(D) Forward Voltage Drop 179004

Right Answer :
Peak Inverse Voltage

Right Option Id : 179002

Question 55

Question Id : 180

Which flag in the 8051 microcontroller is used to indicate carry-out during addition/subtraction and is also set in certain Boolean operations?

Answer :

Option Id

(A) Overflow Flag 180001
(B) Parity Flag 180002
(C) Sign Flag 180003
(D) Carry Flag 180004

Right Answer :
Carry Flag

Right Option Id : 180004

Question 56

Question Id : 181

What is the attenuation of high-frequency components in flat-top sampling due to the sinc pulse roll-off called?

Answer :

Option Id

(A) Aperture effect — 181001
(B) Aliasing — 181002
(C) Nyquist effect — 181003
(D) Distortion — 181004

Right Answer :
Aperture effect

Right Option Id : 181001

Question 57

Question Id : 182

Which protocol reports errors and provides diagnostic information for unsuccessful packet delivery?

Answer :

Option Id

(A) IGMP — 182001
(B) ICMP — 182002
(C) ARP — 182003
(D) TCP — 182004

Right Answer :
ICMP

Right Option Id : 182002

Question 58

Question Id : 183

Which bits in the Program Status Word (PSW) are used to select the register bank in the 8051 microcontroller?

Answer :

Option Id

(A) RS0 and RS1 in the Status Register — 183001
(B) RS1 and RS2 in the Program Control Register — 183002
(C) PSW.5 and PSW.6 in the PSW Register — 183003
(D) PSW.3 (RS0) and PSW.4 (RS1) in the PSW — 183004

Right Answer :
PSW.3 (RS0) and PSW.4 (RS1) in the PSW

Right Option Id : 183004

Question 59

Question Id : 184

What does Mean Time Between Failure (MTBF) represent in system reliability?

Answer :

Option Id

(A) The duration for system repair after failure — 184001
(B) The time taken to detect a failure — 184002
(C) The frequency of system failures in a given time frame — 184003
(D) The total operational time of a system — 184004

Right Answer :
The frequency of system failures in a given time frame

Right Option Id : 184003

Question 60

Question Id : 185

What is the term for a discrete-time signal that is zero for negative time and has a constant magnitude starting from n=0?

Answer :

Option Id

(A) Unit Step Sequence — 185001
(B) Unit Impulse Sequence — 185002
(C) Sinusoidal Sequence — 185003
(D) Exponential Sequence — 185004

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Right Answer :
Unit Step Sequence

Right Option Id : 185001

Question 61

Question Id : 186

What is the path called when the terrain contour blocks the direct wave path in a mobile radio environment?

Answer :

Option Id

(A) Direct Wave Path — 186001
(B) Line-of-Sight Path — 186002
(C) Clear Path — 186003
(D) Obstructive Path — 186004

Right Answer :
Obstructive Path

Right Option Id : 186004

Question 62

Question Id : 187

Which model uses the hybrid π equivalent circuit for analyzing transistors and is characterized by components whose values are independent of frequency?

Answer :

Option Id

(A) Miller Model — 187001
(B) Giacolleto Model — 187002
(C) Hybrid Model — 187003
(D) Thevenin Model — 187004

Right Answer :
Giacolleto Model

Right Option Id : 187002

Question 63

Question Id : 188

What is the correlation called when a signal is compared with its own shifted version to analyze its repetitive patterns?

Answer :

Option Id

(A) Cross Correlation — 188001
(B) Auto Correlation — 188002
(C) Signal Matching — 188003
(D) Phase Comparison — 188004

Right Answer :
Auto Correlation

Right Option Id : 188002

Question 64

Question Id : 189

In the time shifting operation of a discrete-time signal, represented by the equation y(n) = x(n - k), a positive value of k represents _____.

Answer :

Option Id

(A) Time advance (shift to the left) — 189001
(B) No shift in the signal — 189002
(C) Time delay (shift to the right) — 189003
(D) Time scaling — 189004

Right Answer :
Time delay (shift to the right)

Right Option Id : 189003

Question 65

Question Id : 190

Which of the following types of signals exists for both positive and negative time intervals?

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Answer :

(A) Non-causal signals — 190001
(B) Causal signals — 190002
(C) Anti-causal signals — 190003
(D) Unit step signals — 190004

Right Answer :
Non-causal signals

Right Option Id : 190001

Question 66

Question Id : 191

What is the term used for the ratio of the RMS value of the AC component to the DC value of the component in a rectifier circuit?

Answer : Option Id

(A) Efficiency — 191001
(B) Power Factor — 191002
(C) Load Regulation — 191003
(D) Ripple Factor — 191004

Right Answer :
Ripple Factor

Right Option Id : 191004

Question 67

Question Id : 192

Which of the following types of signals satisfies the condition x(n) = x(-n) for all n?

Answer : Option Id

(A) Even (symmetric) signal — 192001
(B) Odd (antisymmetric) signal — 192002
(C) Causal signal — 192003
(D) Non-causal signal — 192004

Right Answer :
Even (symmetric) signal

Right Option Id : 192001

Question 68

Question Id : 193

What is the process of converting continuous-amplitude samples into discrete-time signals called?

Answer : Option Id

(A) Sampling — 193001
(B) Encoding — 193002
(C) Modulation — 193003
(D) Quantization — 193004

Right Answer :
Quantization

Right Option Id : 193004

Question 69

Question Id : 194

What does ARP resolve in a broadcast-based network like Ethernet?

Answer : Option Id

(A) IPv4 addresses to MAC addresses — 194001
(B) MAC addresses to IPv4 addresses — 194002
(C) IPv6 addresses to IPv4 addresses — 194003
(D) MAC addresses to Application Layer addresses — 194004

Right Answer :
IPv4 addresses to MAC addresses

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Right Option Id : 194001

Question 70

Question Id : 195

Which property of a system ensures that a time shift in the input results in a corresponding time shift in the output?

Answer :

Option Id

(A) Shift-invariant system — 195001
(B) Time-varying system — 195002
(C) Linear system — 195003
(D) Non-linear system — 195004

Right Answer :

Shift-invariant system

Right Option Id : 195001

Question 71

Question Id : 196

Which unit in a computer system is represented by a silicon chip capable of performing arithmetic and logical operations?

Answer :

Option Id

(A) Memory — 196001
(B) Timer Unit — 196002
(C) Interrupt Controller — 196003
(D) Microprocessor — 196004

Right Answer :

Microprocessor

Right Option Id : 196004

Question 72

Question Id : 197

Which structure uses separate delays for input and output samples and requires more memory for realization in digital systems?

Answer :

Option Id

(A) Non-canonical structure — 197001
(B) Canonical form — 197002
(C) Direct form-II structure — 197003
(D) State-space representation — 197004

Right Answer :

Canonical form

Right Option Id : 197002

Question 73

Question Id : 198

Which of the following represents the ratio of the distance between the centers of two cells to the radius of the cell in co-channel interference reduction?

Answer :

Option Id

(A) q = D/R — 198001
(B) q = Pt/Pr — 198002
(C) q = A/B — 198003
(D) q = R/D — 198004

Right Answer :

q = D/R

Right Option Id : 198001

Question 74

Question Id : 199

Which coupling method uses a resistor-capacitor combination, where the capacitor allows AC signals to pass while blocking DC components?

Answer :

Option Id

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(A) Direct Coupling
(B) Transformer Coupling
(C) Inductive Coupling
(D) Resistance-Capacitance Coupling

199001
199002
199003
199004

Right Answer :
Resistance-Capacitance Coupling

Right Option Id : 199004

Question 75

Question Id : 200

What is the range of complex variable values called where the Laplace Transform converges?

Answer :

(A) Frequency Spectrum
(B) Region of Convergence (ROC)
(C) Signal Bandwidth
(D) Laplace Domain

Option Id
200001
200002
200003
200004

Right Answer :
Region of Convergence (ROC)

Right Option Id : 200002

Question 76

Question Id : 201

Which of the following structures uses fewer delay elements and introduces an intermediate variable to split the transfer function into poles and zeros?

Answer :

(A) Direct Form-II Structure
(B) Direct Form-I Structure
(C) State-space representation
(D) Cascade form structure

Option Id
201001
201002
201003
201004

Right Answer :
Direct Form-II Structure

Right Option Id : 201001

Question 77

Question Id : 202

Which of the following devices is primarily used for backup storage in computer systems, providing large storage capacity at a low cost?

Answer :

(A) Main memory
(B) Cache memory
(C) Auxiliary memory
(D) Register memory

Option Id
202001
202002
202003
202004

Right Answer :
Auxiliary memory

Right Option Id : 202003

Question 78

Question Id : 203

What is the mathematical expression for the transformer voltage in a half wave rectifier circuit?

Answer :

(A) $v(t) = vm * \cos(\omega t)$
(B) $v(t) = vm * \sin(\omega t)$
(C) $v(t) = vm * e^{(-\omega t)}$
(D) $v(t) = vm * \tan(\omega t)$

Option Id
203001
203002
203003
203004

Right Answer :
$v(t) = vm * \sin(\omega t)$

Right Option Id : 203002

Question 79

Question Id : 204

Which technique involves partitioning memory into modules connected to common memory address and data buses to allow simultaneous access from multiple sources?

Answer :

Option Id

(A) Memory Interleaving — 204001
(B) Pipelining — 204002
(C) Multithreading — 204003
(D) Memory Mapping — 204004

Right Answer :
Memory Interleaving

Right Option Id : 204001

Question 80

Question Id : 205

What is the minimum sampling rate required to avoid aliasing, as per the Nyquist theorem?

Answer :

Option Id

(A) fs = fm — 205001
(B) fs = fm / 2 — 205002
(C) fs = 2fm — 205003
(D) fs = 3fm — 205004

Right Answer :
fs = 2fm

Right Option Id : 205003

Question 81

Question Id : 206

What is ADSL primarily designed for?

Answer :

Option Id

(A) Businesses requiring symmetric data rates — 206001
(B) Residential users with asymmetric data needs — 206002
(C) High-speed corporate networks — 206003
(D) Cable TV internet access — 206004

Right Answer :
Residential users with asymmetric data needs

Right Option Id : 206002

Question 82

Question Id : 207

Which of the following refers to the situation where a read or write operation is performed on the cache, and the requested word is already present in the cache?

Answer :

Option Id

(A) Cache Miss — 207001
(B) Cache Hits — 207002
(C) Cache Overflow — 207003
(D) Cache Write-through — 207004

Right Answer :
Cache Hits

Right Option Id : 207002

Question 83

Question Id : 208

Which silicon chip contains a CPU, RAM, ROM/FLASH, timers, interrupt controllers, and dedicated I/O ports, making it suitable for embedded systems?

Answer :

Option Id

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(A) Microprocessor
(B) Memory Controller
(C) Input/Output Controller
(D) Microcontroller

208001
208002
208003
208004

Right Answer :
Microcontroller

Right Option Id : 208004

Question 84

Question Id : 209

In asynchronous serial transfer, which of the following describes the structure of each character being transmitted?

Answer :

(A) A start bit, character bits, and a stop bit
(B) A parity bit, character bits, and a checksum
(C) Start bit, data bits, and error-correction bits
(D) Data bits, acknowledgement bit, and end-of-transmission signal

Option Id
209001
209002
209003
209004

Right Answer :
A start bit character bits and a stop bit

Right Option Id : 209001

Question 85

Question Id : 210

What does the unit 'Erlangs' measure in telecommunications?

Answer :

(A) The time utilization of a single or multiple channels
(B) The signal strength in a communication channel
(C) The frequency spectrum of the signal
(D) The data transfer rate of a channel

Option Id
210001
210002
210003
210004

Right Answer :
The time utilization of a single or multiple channels

Right Option Id : 210001

Question 86

Question Id : 211

What term refers to connecting amplifier stages directly without DC isolation, typically used when the load is connected in series?

Answer :

(A) Impedance Coupling
(B) Resistance-Capacitance Coupling
(C) Direct Coupling
(D) Transformer Coupling

Option Id
211001
211002
211003
211004

Right Answer :
Direct Coupling

Right Option Id : 211003

Question 87

Question Id : 212

What is the mathematical expression for the even component of a signal?

Answer :

Option Id

(A) x_even(t) = x(t) * x(-t) — 212001

(B) x_even(t) = 1/2 [x(t) + x(-t)] — 212002

(C) x_even(t) = x(t) - x(-t) — 212003

(D) x_even(t) = x(t) + x(-t) — 212004

Right Answer :

x_even(t) = 1/2 [x(t) + x(-t)]

Right Option Id : 212002

Question 88

Question Id : 213

Which method of asynchronous data transfer uses a single control line to time each transfer?

Answer :

Option Id

(A) Handshaking — 213001

(B) Parity control — 213002

(C) Clock synchronization — 213003

(D) Strobe Control Method — 213004

Right Answer :

Strobe Control Method

Right Option Id : 213004

Question 89

Question Id : 214

Which of the following methods involves the source unit generating a "data valid" signal to initiate data transfer and the destination unit generating a "data accepted" signal after accepting the data?

Answer :

Option Id

(A) Source-initiated handshaking — 214001

(B) Clock synchronization — 214002

(C) Strobe control — 214003

(D) Parallel data transfer — 214004

Right Answer :

Source-initiated handshaking

Right Option Id : 214001

Question 90

Question Id : 215

What is the formula for the discharging time constant of a capacitor in a half-wave rectifier with a capacitor filter?

Answer :

Option Id

(A) Td = C / V — 215001

(B) Td = C * R_L — 215002

(C) Td = C + R_L — 215003

(D) Td = C * V — 215004

Right Answer :

Td = C * R_L

Right Option Id : 215002

Question 91

Question Id : 216

Which of the following DMA transfer modes requires the DREQ signal to remain active throughout the operation, with the transfer continuing when the DREQ signal goes high again?

Answer :

Option Id

(A) Hidden mode — 216001

(B) Demand transfer mode — 216002

(C) Block transfer mode — 216003

(D) Cycle stealing mode — 216004

Right Answer :
Demand transfer mode

Right Option Id : 216002

Question 92

Question Id : 217

What is the minimum sampling rate required to recover a band-pass signal with a bandwidth of 2fm?

Answer :

Option Id

(A) fs = 2fm — 217001
(B) fs = 4fm — 217002
(C) fs = fm — 217003
(D) fs = fm/2 — 217004

Right Answer :
fs = 4fm

Right Option Id : 217002

Question 93

Question Id : 218

What transmission media handles communication between offices using multiplexing?

Answer :

Option Id

(A) Switches — 218001
(B) Trunks — 218002
(C) Routers — 218003
(D) Hubs — 218004

Right Answer :
Trunks

Right Option Id : 218002

Question 94

Question Id : 219

In control systems, what is defined as the deviation of the response at peak time from the final value of the response?

Answer :

Option Id

(A) Peak overshoot — 219001
(B) Settling time — 219002
(C) Rise time — 219003
(D) Steady-state error — 219004

Right Answer :
Peak overshoot

Right Option Id : 219001

Question 95

Question Id : 220

Which processor architecture uses a single bus to fetch both instructions and data from a common memory?

Answer :

Option Id

(A) Von-Neumann — 220001
(B) Harvard — 220002
(C) RISC — 220003
(D) CISC — 220004

Right Answer :
Von-Neumann

Right Option Id : 220001

Question 96

Question Id : 221

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Which of the following terms refers to the deviation of the output of a control system from the desired response during steady state?

Answer :

Option Id

(A) Rise time — 221001
(B) Peak overshoot — 221002
(C) Steady state error — 221003
(D) Settling time — 221004

Right Answer :
Steady state error

Right Option Id : 221003

Question 97

Question Id : 222

What does 'load' measure in a trunked radio system?

Answer :

Option Id

(A) The probability of a call being delayed — 222001
(B) The number of active calls in the system — 222002
(C) The traffic intensity across the entire system, measured in Erlangs — 222003
(D) The call drop rate in the system — 222004

Right Answer :
The traffic intensity across the entire system measured in Erlangs

Right Option Id : 222003

Question 98

Question Id : 223

What component is connected in parallel to the emitter resistor to provide a low reactance path for the amplified AC signal and prevent voltage feedback to the input?

Answer :

Option Id

(A) Emitter By-pass Capacitor (C_e) — 223001
(B) Bypass Resistor — 223002
(C) Coupling Capacitor — 223003
(D) Feedback Capacitor — 223004

Right Answer :
Emitter By-pass Capacitor (C_e)

Right Option Id : 223001

Question 99

Question Id : 224

What is the mathematical condition for the periodicity of a discrete signal?

Answer :

Option Id

(A) x[n ± mN] = x(n) — 224001
(B) x[n ± mT] = x(n) — 224002
(C) x[n + N] = x(n) - N — 224003
(D) x[n] = x(n) + mT — 224004

Right Answer :
x[n ± mN] = x(n)

Right Option Id : 224001

Question 100

Question Id : 225

Which type of system is described as having a constant amplitude and frequency of oscillations for bounded input, and is stable when its poles are located on the imaginary axis?

Answer :

Option Id

(A) Marginally Stable System — 225001
(B) Stable System — 225002
(C) Unstable System — 225003
(D) Asymptotically Stable System — 225004

Right Answer :
Marginally Stable System

Right Option Id : 225001

Question 101

Question Id : 226

Which of the following stability criteria provides both a necessary condition and a sufficient condition to determine the stability of a control system?

Answer : Option Id

(A) Routh-Hurwitz Stability Criterion 226001
(B) Nyquist Criterion 226002
(C) Bode Stability Criterion 226003
(D) Root Locus Criterion 226004

Right Answer :
Routh-Hurwitz Stability Criterion

Right Option Id : 226001

Question 102

Question Id : 227

What is the name of the oscillator circuit that uses an amplifier and an LC circuit as a feedback mechanism?

Answer : Option Id

(A) Colpitts Oscillator 227001
(B) Hartley Oscillator 227002
(C) Wien Bridge Oscillator 227003
(D) Crystal Oscillator 227004

Right Answer :
Hartley Oscillator

Right Option Id : 227002

Question 103

Question Id : 228

Which method allows for determining the stability of a control system without calculating the roots of the characteristic equation, by using the sign changes in the first column of the Routh table?

Answer : Option Id

(A) Routh array method 228001
(B) Root locus method 228002
(C) Bode plot method 228003
(D) Nyquist criterion 228004

Right Answer :
Routh array method

Right Option Id : 228001

Question 104

Question Id : 229

What is the process of compressing a signal before quantization and expanding it after reception called?

Answer : Option Id

(A) Quantization 229001
(B) Companding 229002
(C) Encoding 229003
(D) Modulation 229004

Right Answer :
Companding

Right Option Id : 229002

Question 105

Question Id : 230

What does the receiver send when it receives a damaged or duplicate frame?

Answer :

Option Id

(A) ACK — 230001
(B) NACK — 230002
(C) SYN — 230003
(D) FIN — 230004

Right Answer :
NACK

Right Option Id : 230002

Question 106

Question Id : 231

Which technique uses an open-loop transfer function to determine the stability of the closed-loop control system by observing the path of the closed-loop poles?

Answer :

Option Id

(A) Root locus technique — 231001
(B) Bode plot method — 231002
(C) Nyquist criterion — 231003
(D) Routh-Hurwitz criterion — 231004

Right Answer :
Root locus technique

Right Option Id : 231001

Question 107

Question Id : 232

Which type of logic devices allow users to define and modify their functionality after manufacturing?

Answer :

Option Id

(A) Fixed Logic Devices — 232001
(B) Hardwired Logic Devices — 232002
(C) Standard Logic Devices — 232003
(D) Programmable Logic Devices (PLDs) — 232004

Right Answer :
Programmable Logic Devices (PLDs)

Right Option Id : 232004

Question 108

Question Id : 233

Which conditions are used to identify the points on the root locus branches, where the angle condition ensures the angle of the open-loop transfer function is an odd multiple of 180°?

Answer :

Option Id

(A) Stability Condition and Phase Condition — 233001
(B) Phase Condition and Gain Condition — 233002
(C) Root Locus Condition and Frequency Condition — 233003
(D) Angle Condition and Magnitude Condition — 233004

Right Answer :
Angle Condition and Magnitude Condition

Right Option Id : 233004

Question 109

Question Id : 234

Which term refers to the scheme that allocates spectrum efficiency in real-time and adjusts cell sites based on traffic conditions?

Answer :

Option Id

(A) Static Cell Allocation — 234001
(B) Frequency Reuse — 234002
(C) Dynamic Splitting — 234003
(D) Fixed Spectrum Allocation — 234004

Right Answer :
Dynamic Splitting

Right Option Id : 234003

Question 110

Question Id : 235

What term refers to a circuit consisting of two emitter follower stages in cascade with infinite emitter resistance in the first stage?

Answer :

Option Id

(A) Differential Amplifier — 235001
(B) Darlington Pair — 235002
(C) Common-Emitter Amplifier — 235003
(D) Voltage Follower — 235004

Right Answer :
Darlington Pair

Right Option Id : 235002

Question 111

Question Id : 236

What is the condition for a system to be BIBO stable?

Answer :

Option Id

(A) $|y| \leq |x|$ — 236001
(B) $|y| \leq k2$ for $|x| \leq k$ — 236002
(C) $|y| > k2$ for $|x| \leq k$ — 236003
(D) $|y| \geq k2 + k$ — 236004

Right Answer :
$|y| \leq k2$ for $|x| \leq k$

Right Option Id : 236002

Question 112

Question Id : 237

Under which condition is a two-port network considered reciprocal, where the ratio of the output response variable to the input excitation variable remains the same when the excitation and response ports are interchanged?

Answer :

Option Id

(A) Symmetry Condition — 237001
(B) Reciprocity Condition — 237002
(C) Voltage Matching Condition — 237003
(D) Current Condition — 237004

Right Answer :
Reciprocity Condition

Right Option Id : 237002

Question 113

Question Id : 238

A direct route to obtaining the characteristic equation from the general first-order differential equation a(df/dt) + bf = 0 results in which of the following equations?

Answer :

Option Id

(A) $as + b = 0$ — 238001
(B) $as^2 + b = 0$ — 238002
(C) $a + b = 0$ — 238003
(D) $a^{2s} + b = 0$ — 238004

Right Answer :
$as + b = 0$

Right Option Id : 238001

Question 114

Question Id : 239

What are the majority charge carriers in an N-channel JFET?

Answer :

Option Id

(A) Holes — 239001
(B) Electrons — 239002
(C) Ions — 239003
(D) Protons — 239004

Right Answer :
Electrons

Right Option Id : 239002

Question 115

Question Id : 240

What term is represented by ω_0 in the equation s1, s2 = -R/(2L) ± √((R/(2L))^2 - 1/LC) = -α ± √(α^2 - ω_0^2), where α = R/(2L) and ω_0 = 1/√(LC)?

Answer :

Option Id

(A) Damping coefficient — 240001
(B) Natural frequency — 240002
(C) Resonant frequency — 240003
(D) Exponential decay — 240004

Right Answer :
Resonant frequency

Right Option Id : 240003

Question 116

Question Id : 241

What is the process of representing digital 1s and 0s on a transmission link called?

Answer :

Option Id

(A) Modulation — 241001
(B) Companding — 241002
(C) Line Coding — 241003
(D) Quantization — 241004

Right Answer :
Line Coding

Right Option Id : 241003

Question 117

Question Id : 242

In the Go-Back-N protocol, what happens when the receiver gets a sequence number it is not expecting?

Answer :

Option Id

(A) It sends an ACK for the received sequence number. — 242001
(B) It processes the out-of-order sequence silently. — 242002
(C) It silently ignores the unexpected sequence number. — 242003
(D) It requests a retransmission of the unexpected sequence. — 242004

Right Answer :
It silently ignores the unexpected sequence number.

Right Option Id : 242003

Question 118

Question Id : 243

What term describes how well a series RLC circuit responds to its resonant frequency while rejecting all other frequencies, and is directly proportional to the Q factor?

Answer :

Option Id

(A) Selectivity — 243001
(B) Bandwidth — 243002
(C) Quality factor — 243003

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(D) Resonance

Right Answer :
Selectivity

Right Option Id : 243001

Question 119

Question Id : 244

What type of programmable logic device combines features of both PALs and FPGAs, offering up to about 10,000 gates?

Answer :

Option Id

(A) FPGA — 244001
(B) PAL — 244002
(C) ASIC — 244003
(D) CPLD — 244004

Right Answer :
CPLD

Right Option Id : 244004

Question 120

Question Id : 245

In the context of the parallel RLC circuit, what is the term for the reciprocal of impedance, which is related to the frequency at which resonance occurs?

Answer :

Option Id

(A) Impedance — 245001
(B) Reactance — 245002
(C) Resistance — 245003
(D) Admittance — 245004

Right Answer :
Admittance

Right Option Id : 245004

Question 121

Question Id : 246

Which technique improves capacity by reducing the D/R ratio and increasing frequency reuse, while maintaining the cell radius unchanged?

Answer :

Option Id

(A) Cell Splitting — 246001
(B) Frequency Reuse — 246002
(C) Sectoring — 246003
(D) Dynamic Splitting — 246004

Right Answer :
Sectoring

Right Option Id : 246003

Question 122

Question Id : 247

What type of power amplifier operates with the collector current flowing for less than half of the input signal's cycle?

Answer :

Option Id

(A) Class C Power Amplifier — 247001
(B) Class A Power Amplifier — 247002
(C) Class B Power Amplifier — 247003
(D) Class AB Power Amplifier — 247004

Right Answer :
Class C Power Amplifier

Right Option Id : 247001

Question 123

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What is the system function H(s) defined as in a relaxed LTI system?

Answer :

Option Id

(A) H(s) = X(s) / Y(s) — 248001

(B) H(s) = Y(s) / X(s) — 248002

(C) H(s) = X(s) * Y(s) — 248003

(D) H(s) = Y(s) + X(s) — 248004

Right Answer :

H(s) = Y(s) / X(s)

Right Option Id : 248002

Question 124

Question Id : 249

Which law states that the line integral of the magnetic field intensity H around a closed magnetic path is equal to the total current enclosed by the path?

Answer :

Option Id

(A) Ampere's Law — 249001

(B) Fleming's Right Hand Rule — 249002

(C) Ohm's Law — 249003

(D) Faraday's Law — 249004

Right Answer :

Ampere's Law

Right Option Id : 249001

Question 125

Question Id : 250

Which rule is used to determine the direction of magnetic flux in a current-carrying conductor, where the curled fingers show the direction of current and the thumb indicates the direction of flux flow?

Answer :

Option Id

(A) Right hand curl rule — 250001

(B) Fleming's Left Hand Rule — 250002

(C) Ampere's Rule — 250003

(D) Lenz's Law — 250004

Right Answer :

Right hand curl rule

Right Option Id : 250001